

Kulan Perera

Smart Waste Management System for Sustainable Cities

 Undergraduate Research Paper Checking

Document Details

Submission ID

trn:oid::3618:135532775

Submission Date

Apr 17, 2026, 1:31 PM GMT+5:30

Download Date

Apr 17, 2026, 1:36 PM GMT+5:30

File Name

Smart Waste Management System for Sustainable Cities .pdf

File Size

438.5 KB

8 Pages

4,773 Words

29,899 Characters

5% Overall Similarity

The combined total of all matches, including overlapping sources, for each database.

Filtered from the Report

- Bibliography
- Quoted Text
- Small Matches (less than 8 words)

Match Groups

-  **14 Not Cited or Quoted 4%**
Matches with neither in-text citation nor quotation marks
-  **4 Missing Quotations 1%**
Matches that are still very similar to source material
-  **0 Missing Citation 0%**
Matches that have quotation marks, but no in-text citation
-  **0 Cited and Quoted 0%**
Matches with in-text citation present, but no quotation marks

Top Sources

- 4%  Internet sources
- 3%  Publications
- 4%  Submitted works (Student Papers)

Match Groups

- 14 Not Cited or Quoted** 4%
Matches with neither in-text citation nor quotation marks
- 4 Missing Quotations** 1%
Matches that are still very similar to source material
- 0 Missing Citation** 0%
Matches that have quotation marks, but no in-text citation
- 0 Cited and Quoted** 0%
Matches with in-text citation present, but no quotation marks

Top Sources

- 4% Internet sources
- 3% Publications
- 4% Submitted works (Student Papers)

Top Sources

The sources with the highest number of matches within the submission. Overlapping sources will not be displayed.

1	Internet	icitr.uom.lk	2%
2	Publication	S.R. Liyanage, N.S.N. Basnayake, M.G.U.D. Gamage, P.A.N. Prabhashi, Sanvitha Ka...	<1%
3	Internet	arxiv.org	<1%
4	Internet	www.mdpi.com	<1%
5	Internet	www.preprints.org	<1%
6	Student papers	University of Westminster on 2026-04-17	<1%
7	Internet	vbn.aau.dk	<1%
8	Student papers	University of Nottingham on 2026-04-08	<1%
9	Student papers	Teaching and Learning with Technology on 2024-06-21	<1%
10	Student papers	University of Strathclyde on 2026-03-30	<1%

11	Publication	Choi, Hyungu. "Predictive Modeling of Aircraft Arrival Times in the Terminal Man...	<1%
12	Student papers	The Robert Gordon University on 2026-04-16	<1%
13	Publication	Yutaka Sugihara, Aleksandar Milosavljevic, Skaidre Jankovskaja, Magnus Falk. "A ...	<1%
14	Internet	conf.papercept.net	<1%
15	Internet	theses.hal.science	<1%

Smart Waste Management System for Sustainable Cities

K.J.I Perera

dept. of information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22891754@my.sliit.lk

O.M.K.T. Perera

dept. of information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22333698@my.sliit.lk

B.N.P Galwatta

dept. of information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22358202@my.sliit.lk

B.T.D.Frenando

dept. of information Technology
Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
it22167446@my.sliit.lk

Mahima Weerasinghe

Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
mahima.w@sliit.lk

Lakmini Abeywardhana

Sri Lanka Institute of Information
Technology
Malabe, Sri Lanka
lakmini.d@sliit.lk

Abstract - Rapid urbanization and changing consumption patterns are increasing the volume and complexity of municipal solid waste, straining conventional collection and disposal practices, and contributing to bin overflow, illegal dumping, and low recycling rates. This paper proposes an integrated smart waste management framework that combines multimodal AI for disposal guidance, a Pay-As-You-Throw-inspired fee mechanism informed by LSTM based waste prediction and XGBoost based tax regression, a bilingual (English/Sinhala) RAG based conversational assistant over municipal documents, and a temporal LSTM based forecasting framework for ward-level waste demand prediction, integrated with resource allocation and route optimization for efficient municipal waste collection. . Experiments on synthetic and real world inspired datasets show strong waste-classification performance, accurate weekly waste prediction, improved response relevance in the conversational agent, and superior forecasting accuracy relative to a purely temporal LSTM baseline, indicating the potential of the proposed system to support sustainable, data-driven urban waste management.

Keywords—Smart Waste Management, Multimodal AI, Behavior Based Taxation, Conversational Agents, Municipal Solid Waste, LSTM, XGBoost.

INTRODUCTION

Municipal solid waste (MSW) management is a critical public service that directly affects environmental quality, public health, and citizens' quality of life in urban areas. In rapidly growing cities, including those in Sri Lanka, rising waste volumes and diversified waste streams have outpaced the capabilities of conventional systems that rely

on static collection schedules, manually configured routes, and limited monitoring of bin conditions. Fixed route collection often results in unnecessary trips to underfilled bins, overflow at high demand locations, and redundant fuel and labor. expenditures. At the same time, limited visibility into real-time operations makes it difficult for municipal authorities to plan proactively or to react quickly to localized surges in demand.

At the household level, a lack of clear, accessible guidance on waste separation leads to persistent misclassification among the organic, recyclable, and residual fractions. Citizens may be unaware of appropriate disposal techniques for specific items or may lack confidence about which bin to use and whether a particular bin has sufficient remaining capacity to accept their waste. This situation is compounded by limited enforcement and the absence of transparent, behavior linked financial mechanisms. Flat-rate taxation or simple volume-based fees fail to capture individual household behavior and often provide little incentive to reduce non-recyclable waste or to improve segregation quality. Consequently, recyclable streams become contaminated, landfill dependency remains high, and illegal dumping emerges as a recurring issue in marginal or underserved areas.

Recent advances in artificial intelligence, mobile computing, and sensing technologies offer new opportunities to address these challenges. Multimodal deep learning models can classify waste from images or text descriptions, while gradient boosting algorithms can model complex decision logic from tabular data. Time-series

5 models such as Long Short-Term Memory (LSTM) networks enable the prediction of household waste generation, and Temporal LSTM models can capture time-dependent waste generation patterns across wards for city-scale waste demand forecasting. When grounded in official documentation, large language models can provide citizens with accurate and localized guidance at scale.

4 This research presents a smart waste management system designed as a comprehensive, end-to-end platform that couples these advances with practical interfaces for citizens, collectors, and planners. The work is organized into four components (objectives):

- AI guided disposal support spanning waste identification, bin capacity checking, and bin-overflow forecasting.
- A household fee model using LSTM and XGBoost to link waste generation patterns to fair, explainable taxation and incentives, delivered through mobile applications.
- 6 • A bilingual (English/Sinhala) conversational agent using a Retrieval-Augmented Generation (RAG) approach over municipal documents to provide localized waste information.
- 14 • A temporal forecasting pipeline using LSTM models that integrates municipal operational data to predict ward level waste demand and supports demand scoring, resource allocation, and route optimization for efficient collection planning.

RELATED-WORK

Existing work commonly represents smart, household-facing waste support in three streams: automated waste classification, IoT-based bin monitoring with collection optimization, and time-series prediction of waste generation and bin fill levels. In many systems, computer-vision or text-based models are used to identify the waste category to support correct separation. [1]. However, these approaches often stop at category prediction and do not provide user-specific technique selection (e.g., reuse/recycle/upcycle) together with practical, step-by-step disposal guidance. Another widely used direction equips bins with sensors and connectivity to estimate fill levels and enable more efficient disposal planning. [2], but such solutions are frequently designed from the municipal operations perspective and provide limited assistance at the moment of disposal. Forecasting oriented studies use temporal models (including recurrent neural networks) to predict future waste generation or bin status to support proactive planning. [3], yet purely temporal predictors may ignore spatial interactions between areas that influence localized overflow risk. Building on these common

foundations, this combines multimodal identification with user-centered guidance and operational capacity/overflow prediction to support correct disposal decisions better and reduce overflow events.

In parallel, the economic research on municipal waste management has promoted Pay-As-You-Throw (PAYT) and similar pricing systems based on volume or weight to encourage waste reduction and better source separation. [4], [5], [6]. In practice, these programs usually operate with fixed tariff schedules based on observed waste amounts or standard container sizes. [5], [6]. While these designs can lower residual waste, typical PAYT programs seldom use predictive analytics or targeted behavioral segmentation. [4], [5]. As a result, short-term changes and ongoing high-waste behavior may be charged the same, and historical data may not be fully used when setting tariffs that are fair and responsive. [4].

Conversational systems for waste management typically take the form of static FAQ portals or rule-based chatbots maintained by municipal authorities. These systems can provide basic information about collection schedules and allowed materials, but are often limited in conversational flexibility and language coverage. [7]. With the emergence of large language models (LLMs), interest has grown in deploying AI-based assistants for public services. [8]. However, in the absence of robust grounding in official documents and strict constraints, such systems may generate incorrect or non-compliant information. This risk is particularly salient in domains where local regulations, bin color codes, and fee rules vary by jurisdiction and must be communicated precisely in local languages.

Spatio temporal modeling is essential for forecasting zone-level waste demand in municipal systems. In the EcoFit framework, temporal patterns in historical waste data are captured using Long Short-Term Memory (LSTM) networks, which effectively model seasonal trends and short-term fluctuations in municipal waste generation. These temporal predictions are derived from multiple municipal data sources, including service logs, complaint records, operational indicators, and environmental factors, enabling more reliable demand estimation across wards. Similar temporal forecasting approaches have demonstrated strong performance in urban demand prediction tasks such as traffic and city service management. [9], [10]. Based on the predicted demand risk levels, the system generates a prioritized operational plan that supports demand scoring, resource allocation, and route optimization for municipal waste collection. By translating predictive insights into actionable service planning, the framework facilitates data-driven resource

allocation and operational decision-making for municipalities. [11].

The framework proposed in this paper extends prior work by tightly integrating multimodal AI for household guidance, predictive behavior-based tariffs, document-grounded conversational assistance, and LSTM based waste demand forecasting, combined with operational planning mechanisms such as demand scoring, resource allocation, and route optimization into a unified architecture. It explicitly targets both the citizen-facing and municipal-planning aspects of waste management, with an emphasis on explainability, localization, and operational applicability.

METHODOLOGY

A. System Architecture Overview

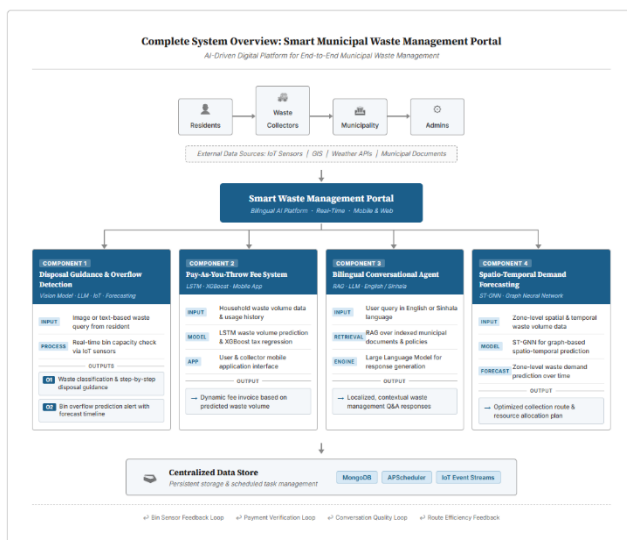


Figure 1. System Architecture Diagram

The overall architecture is designed as a multi-layer framework that links data acquisition, machine learning models, and stakeholder facing applications. It integrates household interaction data, IoT bin telemetry (distance-based fill measurements), municipal records (routes, schedules, policies, and zone boundaries), and contextual factors that influence waste generation. These inputs feed a model layer comprising specialized modules: a multimodal recognition and guidance pipeline that classifies waste from image or text, predicts a user-specific technique (reuse/recycle/upcycle), checks current bin capacity, and forecasts overflow risk; a temporal prediction module based on LSTM to estimate weekly household waste generation by category, coupled with an XGBoost regressor that maps observed or predicted waste quantities to behavior-linked fee values; a bilingual conversational

assistant implemented with retrieval-augmented generation (RAG) using hybrid retrieval over a vector store to ground responses in municipal documents; and a city-scale forecasting module based on a on temporal LSTM networks that model historical waste generation patterns across municipal wards to support demand prediction, resource allocation, and route optimization. The application layer operationalizes these capabilities through a resident mobile app (submission, guidance, bin-fit decisions, tax simulation, and bilingual queries), a collector app (verification and tariff administration), and a web-based planning dashboard (visualization of zone-level forecasts and allocation scenarios).

B. Datasets

The study uses a mixture of real and synthetic datasets to reflect typical municipal data constraints. A synthetic household-level dataset is derived from aggregated truck logs and enriched with seasonal, event-based, and demographic signals. The disposal guidance datasets comprise labeled and augmented images of common household waste items, paired with a text dataset of user-like descriptions of waste. A Google Forms survey dataset captures household awareness, recycling habits, and stated preferences, with labels indicating preferred techniques (reuse, recycle, upcycle). Operational monitoring data comprise per-bin IoT sensor readings (e.g., ultrasonic distance/height) maintained as daily fill-level time series after calibration. City-scale planning data include zone boundaries, historical zone-level waste loads, population estimates, and indicators for public holidays and events. Finally, official municipal PDF documents provide the knowledge base for citizen guidance, supplemented by a Sinhala instruction dataset curated for language-model adaptation.

C. Integrated System Components

1. End-to-End Disposal Guidance with Capacity Checking and Overflow Forecasting

This component provides end-to-end disposal guidance, from waste identification to bin-capacity assessment and bin overflow forecasting. The workflow is initiated when a household submits either an image of a waste item or a short text description through the user application, and may optionally include an approximate volume estimate.

For image based identification, the input is resized and normalized, visual features are extracted using a convolutional backbone, and a classification head produces a probability distribution over the municipality-defined waste categories. For text only input, the description is tokenized and embedded, and a sequence model generates

a representation for classification. Both modalities share a common label space so that image and text inputs are handled consistently. After categorization, the system predicts whether a user will reuse, recycle, or upcycle a technique based on the predicted waste category and the user's context. [12]. Using the selected technique, the system recommends an appropriate disposal method and generates an actionable workflow with step-by-step guidance on how to implement it, including safety notes and practical tips. The technique prediction is produced by a supervised model that maps the category and user context to the most suitable option.

In parallel, the system evaluates whether the selected smart bin can accommodate the proposed deposit. Each bin records a daily sensor measurement of the distance from the sensor to the waste surface, which is converted to an estimated fill level and remaining capacity using the bin geometry. [13], [14]. When a user provides a volume estimate, the system compares the estimate with the remaining capacity and returns a bin-fit decision, which is used to confirm the bin or to recommend alternative actions.

Beyond instantaneous capacity checks, the system predicts future overflow using a lag-based time-series forecasting approach. [15]. For each household bin, the daily sensor time series is maintained, normalized to a fill percentage, and used to forecast fill levels over a planning horizon. The forecasting model is periodically refit and identifies the date at which the predicted fill level exceeds a predefined overflow threshold. Overall, the component supports correct household disposal decisions while providing municipalities with actionable capacity and forecast signals to reduce overflow risk across the bin network.

2. Household Waste-Based Smart Tax or Fee Calculation System

This component proposes a transparent, behavior-based fee model within a hybrid AI deterministic framework by coupling time-series forecasting of household waste with a gradient-boosting taxation module. Prior work indicates that machine-learning and deep-learning methods can achieve strong performance in municipal solid-waste forecasting and in learning temporal patterns, which motivates their use here. [16], [17].

In the absence of household-level measurements, aggregated weekly waste totals are disaggregated into realistic household records via a simulation pipeline. The resulting synthetic dataset supports classifying households by waste-generation behavior and applying corresponding multipliers or discounts. Such simulation-based

reconstruction is appropriate when fine-grained household observations are not feasible. [18], and weight-based charging is widely regarded as a fair basis for municipal taxation [19].

Weekly waste quantities are forecast using an LSTM model trained on historical sequences; the model also supports missing value imputation and scenario-based analyses, while hybrid ML strategies can improve robustness. [16], [17]. Tax values are then estimated with XGBoost, leveraging the strong empirical performance of gradient boosting on environmental tabular datasets. [20].

Final charges are computed via a deterministic billing formula to ensure that the resulting taxes are fair, reproducible, and auditable. The user-facing mobile application enables waste reporting, tax monitoring, and visibility into area rankings; in addition, gamification mechanisms (e.g., leaderboards and rewards) are incorporated to promote sustained engagement and pro-environmental behavior, consistent with prior research. [21]. Finally, the collector application supports weight verification, household record review, and base-rate adjustments to ensure that operational measurements remain aligned with municipal policy objectives.

3. Conversational Agent for Waste Issue Resolution and Service Guidance.

The agentic component provides a conversational interface that connects citizens to accurate, localized waste management information. It follows a Retrieval-Augmented Generation (RAG) pattern to ensure that answers are grounded exclusively in official municipal documents. [22]. The knowledge base is constructed by extracting text from relevant PDFs, cleaning it, and splitting it into segments of suitable length. Each segment is embedded into a vector space using a sentence-level encoder, and a FAISS index is built for similarity search. In parallel, a keyword-based BM25 index is created to support exact matching of location names, collection times, and domain-specific terms in Sinhala.

When the user submits a query, the system detects whether the language is English or Sinhala and then computes an embedding for the query. For Sinhala queries, candidate segments are retrieved using a weighted hybrid approach that combines BM25 keyword scores with FAISS semantic similarity scores, prioritizing keyword matches while retaining semantic recall. The top-ranked segments are included as context in the prompt to the language model. For English queries, FAISS retrieves a set of candidate segments, which are then reranked by a cross-encoder

model that scores the semantic relevance of each query segment pair.

The top ranked segments are included as context in the prompt to the language model. Retrieved segments are filtered based on the user's registered geographic location to ensure that location-specific schedules and services are returned. For English queries, responses are generated using the LLaMA 3.1 8B model. For Sinhala queries, a QLoRA tuned Gemma model is used, trained on Sinhala municipal content and typical citizen queries [23]. The prompt instructs the model to answer strictly based on the provided context and to avoid inventing unsupported rules or policies. The conversational agent can respond to questions about collection days, segregation instructions, complaint procedures, and other waste related topics, tailored to the user's geographic location. This reduces the volume of routine inquiries handled by municipal staff and improves citizens' access to consistent, up-to-date guidance [24].

4. Urban Service Demand Forecasting and Resource Optimization Using Temporal Machine Learning

The proposed component addresses city-level waste management planning by forecasting ward-level demand and translating these forecasts into operational resource allocation strategies within the Colombo Municipal Council framework. The system integrates citizen complaint records, municipal service logs, and hourly weather observations to model dynamic patterns of waste accumulation and service performance across all wards. A temporal modeling approach based on Long Short-Term Memory (LSTM) networks is employed to capture historical behavioral trends within each ward. The temporal input features include hourly complaint counts, unresolved complaint ratios, service collection outcomes (collected, partial, missed), estimated waste volume, bin clearance statistics, and precipitation-related variables.

This enables the model to learn seasonal variations, short-term demand fluctuations, and environmental factors affecting waste accumulation. By analyzing these time-dependent patterns, the LSTM model generates ward-level demand predictions that indicate potential waste accumulation risks in the near future. The predicted outputs are expressed as risk categories representing different levels of service urgency. The predicted risk levels are subsequently transformed into actionable operational insights through a demand scoring mechanism that combines the predicted risk scores with additional operational indicators such as complaint volume, missed services, and overflow reports. This scoring mechanism

allows municipal planners to prioritize wards requiring immediate attention.

Based on these prioritized demand scores, the system performs resource allocation, distributing available municipal crews and collection resources across wards according to urgency levels and workload balancing considerations. To further support operational planning, a route optimization module generates efficient collection routes for each crew. The routing process considers depot starting locations, ward priorities, and estimated travel distances to minimize unnecessary travel while ensuring that high-priority wards are serviced earlier.

A web-based municipal planning dashboard visualizes the dynamic ward risk map, prioritized service queues, and recommended resource deployment strategies. This interface converts complex model outputs into interpretable decision support tools, enabling data driven waste management planning and improving operational efficiency in an urban municipal context.

RESULTS

The integrated system was evaluated module-wise to assess the feasibility and effectiveness of the proposed architecture. Overall, the results indicate robust performance across system functions. In the disposal guidance pipeline, the EfficientNetB0 based image classifier showed strong waste category recognition under varied lighting and background conditions, with stable validation behavior and training-validation trends that suggest limited overfitting.

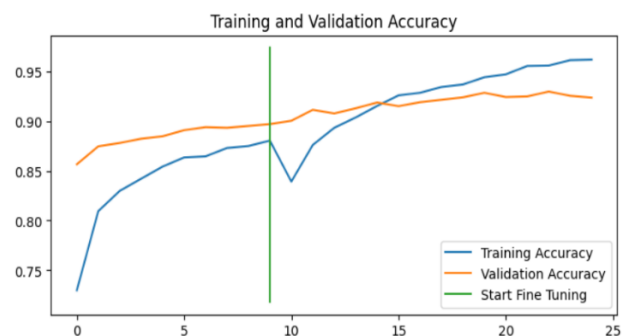


Figure 2. Model Training Accuracy Graph

The text-based classifier also performed well on descriptive queries, with most errors arising from generic or ambiguous inputs. In addition, the XGBoost-based technique recommendation model demonstrated reliable prediction of whether reuse, recycling, or upcycling was most suitable for a given item and user profile. Finally, scenario-based evaluations that coupled classification with

IoT derived bin status suggested fewer simulated overflow events than a non-adaptive baseline

Task/Model	Metric	Value
Text classification (API model)	Accuracy	0.98
Technique prediction (XGBoost)	Accuracy	0.85
Overflow prediction (Random Forest)	MAPE	8.37%

Table 1. Models Metrics Values

In the Smart tax calculation system, the LSTM models accurately predicted weekly household waste. They showed low RMSE and MAE along with high R^2 values. These results captured seasonal patterns and household behaviors, consistent with earlier LSTM based forecasting studies [16], [17]. The XGBoost based tax model matched reference values well and performed well on outliers, supporting the use of gradient boosting for environmental tabular datasets [20]. Overall, the hybrid AI and deterministic framework delivered realistic billing results while keeping a clear link between penalties or discounts and user behavior

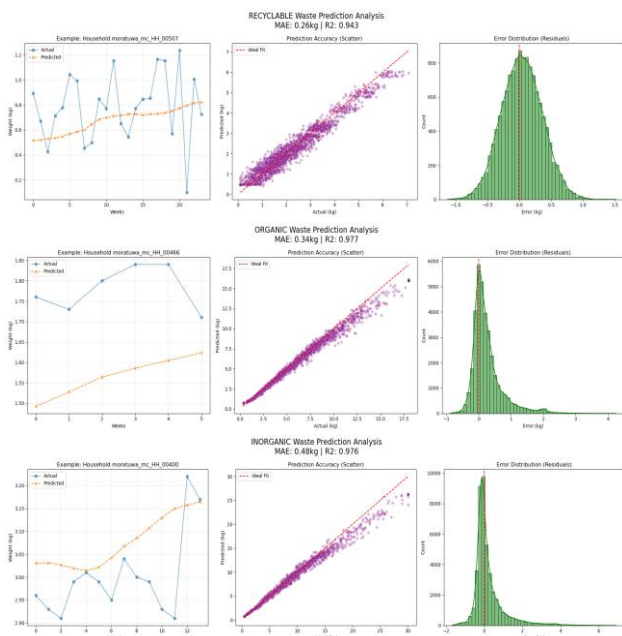


Figure 3. Model Waste Weight Prediction Analysis in three categories

Simulations over several months showed that high-waste households consistently faced higher fees. In contrast, low waste households enjoyed discounts and recognition on the leaderboard. This pattern matches previous gamification research, which suggests that incentives and social

comparison can encourage lasting behavior change. [21].

Model	Regression Metrics	Value
Tax prediction (XGBoost)	R^2	0.9989
	RMSE	Rs. 3.68
	MAE	Rs. 0.78
	MAPE	1.85%

Table 2. Tax Prediction Model Metrics Values

For the conversational assistant, cross encoder re-ranking improved the relevance of retrieved document segments, and a hybrid retrieval strategy that combined keyword based and semantic search performed well for Sinhala queries that included location names and collection schedules. Human evaluation further indicated that responses were accurate and aligned with municipal documents when the relevant context was retrieved, while location based filtering increased precision for schedule related queries by restricting candidates to the user's registered area. Overall, grounding responses in official documents reduced the risk of hallucinated rules, and bilingual support improved accessibility.

Metric	Mean	Min	Max
BLEU	0.635	0.0	1.0
ROUGEL	0.774	0.087	1.0
BERTScore-F1	0.709	0.075	1.0
BLEU	0.635	0.0	1.0

Table 3. English RAG evaluation

The English RAG agent achieves higher lexical and sequence overlap (BLEU = 0.635, ROUGE-L = 0.774), while maintaining reasonable semantic similarity (BERTScore-F1 = 0.709), reflecting good performance on English queries with occasional variations in phrasing.

Metric	Mean	Min	Max
BLEU	0.513	0.112	1.0
ROUGE-L	0.450	0.0	1.0
BERTScore-F1	0.964	0.910	1.0
Metric	Mean	Min	Max

Table 4. Sinhala RAG evaluation

The Sinhala RAG agent demonstrates moderate lexical and sequence overlap with the reference answers (BLEU = 0.513, ROUGE-L = 0.450), while achieving high semantic fidelity (BERTScore-F1 = 0.964), indicating accurate understanding and generation of Sinhala responses.

For city-scale forecasting, the proposed system applies a temporal LSTM based prediction model to capture historical waste generation patterns within each municipal ward. By learning seasonal trends, short-term demand fluctuations, and environmental influences from historical operational data, the model improves the identification of high risk wards that require immediate attention in municipal collection planning. Event driven case analyses further indicated that the model effectively captures sudden increases in service demand caused by factors such as complaint surges, missed collections, or adverse weather conditions, enabling more reliable detection of operational hotspots. These forecasts are integrated into a municipal planning dashboard that supports exploration of alternative resource allocation and route planning scenarios through an intuitive interface for decision-makers.



Figure 4. Model Training Accuracy Graph

Task/Model	Metric	Value
Waste demand prediction (LSTM)	Accuracy	0.781
	Precision (Marco Avg)	0.781
	Recall (Marco Avg)	0.747
	F1-Score (Marco Avg)	0.759
	Best Validation (Marco Avg)	0.830

Table 5. LSTM Waste Demand Prediction Metrics

CONCLUSION

This paper presented an integrated smart waste management framework tailored for sustainable cities that combines multimodal AI, predictive fee modeling, document-grounded conversational assistance, and spatio temporal forecasting. By jointly addressing household behavior, economic incentives, information access, and municipal planning, the system moves beyond siloed solutions and instead offers a holistic digital infrastructure for waste management. The combination of image and text classification, personalized recommendations, and IoT bin monitoring improves segregation and reduces overflow at the household and street levels. The behavior-based fee model, powered by LSTM and XGBoost, links waste-generation patterns to transparent, explainable taxes. At the same time, mobile applications make these mechanisms visible and actionable for both residents and collectors. The RAG-based conversational agent provides accessible, localized, and bilingual guidance grounded in official documents, reducing the informational burden on authorities and increasing policy compliance. The LSTM based forecasting component models temporal waste demand patterns across municipal wards using historical operational data. The predicted risk levels are converted into demand scores to support priority-based resource allocation for municipal collection crews and vehicles. A route optimization mechanism then generates efficient collection routes based on ward priorities and travel distances, supporting data-driven city-scale waste management planning. Together, the results demonstrate how data science and AI can be leveraged to create more responsive, fair, and sustainable municipal waste systems.

FUTURE WORK

Future work will focus on piloting the system in partner municipalities and collecting longitudinal, anonymized operational data to further calibrate models and evaluate fairness impacts. Additional improvements include expanding localized training data to enable more robust classification, strengthening the conversational assistant through improved document updating and citation, and linking demand forecasting to route optimization and traffic-aware scheduling. Establishing standardized interfaces and governance mechanisms will also be important for responsible integration with other smart-city services.

References

- [1] A. Zeb, Q. Ali, M. Q. Saleem, K. M. Awan, A. S. Alowayr, J. Uddin, S. Iqbal and B. Faisal, "A proposed IoT-enabled smart waste bin management system and efficient route selection," *International Journal of Distributed Sensor Networks*, vol. 2019, 2019.
- [2] H. Zhang, H. Cao, Y. Zhou, C. Gu and D. Li, "Hybrid deep learning model for accurate classification of solid waste in the society," *Urban Climate*, vol. 49, 2023.
- [3] M. Helmy and M. S. Hassan, "Smart monitoring of waste bins and municipal solid waste forecast using LSTM for smart cities," *AIP Conference Proceedings*, vol. 3034, no. 1, p. 8, 2024.
- [4] G. Romano and L. Masserini, "Pay-as-you-throw tariff and sustainable urban waste management: An empirical analysis of relevant effects," *Journal of Environmental Management*, vol. 347, 2023.
- [5] V. Elia, M. G. Gnoni and F. Tornese, "Designing Pay-As-You-Throw schemes in municipal waste management services: A holistic approach," *Waste Management*, vol. 44, pp. 188-195, 2015.
- [6] J. E. Aldy and M. L. Miranda, , "Unit pricing of residential municipal solid waste: lessons from nine case study communities," *Journal of Environmental Management*, vol. 52, no. 1, pp. 79-93, 1998.
- [7] R. D. Zota, I. A. Cîmpeanu, D. A. agomir and M. A. Lungu, "Practical Approach for Smart and Circular Cities: Chatbots Used in Waste Recycling," *Applied Sciences*, vol. 14, no. 7, 2024.
- [8] A. S. Rathnayake, T. D. H. Nhat Nguyen and Y. Ahn, "Factors Influencing AI Chatbot Adoption in Government Administration: A Case Study of Sri Lanka's Digital Government," *Administrative Sciences*, vol. 15, no. 5, 2025.
- [9] Y. Liu, C. Shahabi, R. Yu and Y. Li , "Diffusion Convolutional Recurrent Neural Network: Data-Driven Traffic Forecasting," in *International Conference on Learning Representations (ICLR)*, 2018.
- [10] B. Yu, H. Yin and Z. Zhu, "Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting," in *International Joint Conference on Artificial Intelligence (IJCAI)*, Stockholm, Sweden, 2018.
- [11] Y. Xu, Z. Liu and C. Dong, "Machine learning approaches for dynamic resource allocation in smart cities," *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4211-4225, 2023.
- [12] C. A. Info, C. Bauer and E. Zangerle, "Evaluating Recommender Systems: Survey and Framework," *ACM Computing Surveys*, vol. 55, no. 8, pp. 1-38, 2022.
- [13] S. Vishnu, S. R. J. Ramson, S. Senith, T. Anagnostopoulos, A. M. Abu-Mahfouz, X. Fan, S. Srinivasan and A. A. Kirubaraj, "IoT-Enabled Solid Waste Management in Smart Cities," *Smart Cities*, vol. 4, no. 3, pp. 1004-1017, 2021.
- [14] R. S. Podder, M. S. Varkey, J. John, M. Selvi, . N. Sensarma, A. Kannan and S. V. N. S. Kumar, "Smart Prediction and Monitoring of Waste Disposal System Using IoT and Cloud for IoT Based Smart Cities," *Wireless Personal Communications*, vol. 122, no. 1, pp. 243-275, 2021.
- [15] N. A. K. e. al., "AI-enabled smart IoT waste bin monitoring and dynamic collection," in *Int. Conf. on Smart Cities and Green ICT Systems*, Porto, Portugal, 2025.
- [16] J. D. Herath, D. Rathnayake, O. Mudannayake, M. Fernando and . D. K. Fernando, "Exploring Machine Learning and Deep Learning Approaches for Multi-Step Forecasting in Municipal Solid Waste Generation," *IEEE Access*, vol. 10, pp. 570-585, 2022.
- [17] R. Md Noor, I. Ahmedy and F. Fatovatkiah, "Waste Prediction Approach Using Hybrid Long Short-Term Memory with Support Vector Machine," *International Journal of Computational Intelligence Systems* , vol. 17, no. 1, 2024.
- [18] A. Lakhout, M. Shaban, A. Alatawi, S. Y. Abbas, E. Asiri, T. A. Juhni and M. Elsawy, "Machine-learning approaches in geo-environmental engineering: Exploring smart solid waste management," *Journal of Environmental Management*, vol. 330, 2023.
- [19] N. BLAGOEVA and V. GEORGIEVA, "A FAIR TAX BASE FOR THE MUNICIPAL WASTE CHARGE – KEY TO AN EFFECTIVE WASTE MANAGEMENT POLICY IN THE REPUBLIC OF BULGARIA," *Management of Sustainable Development*, vol. 23, no. 2, pp. 247-256, 2023.
- [20] M. A. Olawumi, B. I. Oladapo and R. A. Olawale, "Revolutionising waste management with the impact of Long Short-Term Memory networks on recycling rate predictions," *Waste Management Bulletin*, vol. 2, no. 3, pp. 266-274, 2024.
- [21] V. Tundjungsari, B. P. Dwi Putranto, M. B. Ulum and N. Anwar, "An Integrated Model for Circular Waste Management Using the Internet of Things, Semantic Web, and Gamification (Circonomy): Case Study in Indonesia," *JMIR Serious Games*, vol. 13, 2025.
- [22] M. K. Zafar kizi and Y. Suh, "Design and Performance Evaluation of LLM-Based RAG Pipelines for Chatbot Services in International Student Admissions," *Electronics*, vol. 14, no. 15, 2025.
- [23] P. R. Mouton, L. O. Hall and S. S. Alahmari, "Repeatability of Fine-Tuning Large Language Models Illustrated Using QLoRA," *IEEE Access*, vol. 12, pp. 221-231, 2024.
- [24] A. Androutsopoulou, N. Karacapilidis, . E. Loukis and . Y. Charalabidis, "Transforming the communication between citizens and government through AI-guided chatbots," *Government Information Quarterly*, vol. 36, no. 2, pp. 358-367, 2019.